MATH 180 Strategies of Problem Solving

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November 2, 2026

Outline

1. SoPS Lecture 20
   - Finite Sets
   - Pigeonhole Principle
   - Wrapping up

Review

Last time

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- Relations, Functions

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- Injective, Surjective, Bijective

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- Relations, Functions
- Injective, Surjective, Bijective
- Cardinality, Hilbert's Hotel

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Finite means what?

Question

How should we define a set to be **finite**?

Finite sets

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Definition

A set A is called **finite** if it is empty or there exists a $k \in \mathbb{N}$ for which there is a bijection from $\mathbb{N}_k$ to A.

Example

Problem

Show that $\mathbb{N}_j$ is finite for all j.

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Answer

We should choose $k = j$. That's the only value we should expect to have a bijection!

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There's lots! We just need to demonstrate that there's at least one.

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The **identity function**

$$f : \mathbb{N}_k \to \mathbb{N}_k, \quad f(x) = x$$

in particular is a bijection

Example

Problem

Show that $A = \{\heartsuit, \spadesuit, \diamondsuit, \clubsuit\}$ is finite.

Must show that there's a $k \in \mathbb{N}$ with a bijection from $\mathbb{N}_k$ to A.

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What k should we choose?

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What k should we choose?

Answer

We should choose $k = 4$.

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What bijection from $\mathbb{N}_4$ to A is there?

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What bijection from $\mathbb{N}_4$ to A is there?

Answer

There's lots! We just need to show that there's at least one. For example, $f : \mathbb{N}_4 \to A$ defined by

$$f(1) = \heartsuit, \quad f(2) = \spadesuit, \quad f(3) = \diamondsuit, \quad f(4) = \clubsuit$$

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Therefore f fails to be injective.

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Therefore f fails to be injective.

Pigeonhole Principle

Theorem (Pigeonhole Principle)

Let j and k be integers with $k > j$. If $f : \mathbb{N}_k \to \mathbb{N}_j$, then f is not injective.

Application

Problem

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Fine print:

- we always consider someone to be their own acquaintance

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- nobody knows everyone at the party
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Guests a and b have the same number of acquaintances at the party!

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Problem

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Observation:

- the number p is prime, which might explain why Jenny didn't want to change it.

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For each k, define r_k to be the remainder of n_k divided by p.

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$$f : \mathbb{N} \to \mathbb{N}_k \quad \text{bijection}$$

Use f to define a new function

$$g : \mathbb{N}_{k+1} \to \mathbb{N}_k, \quad g(k) = f(k).$$

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Our original assumption that $\mathbb{N}$ was finite must be false.

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Final thoughts

Reflection

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- Give the formal definition of a finite set.
- Explain why there's a bijection between $\mathbb{N}_j$ and $\mathbb{N}_k$ if and only if $j = k$.

Submit your responses by **Tonight!**